Rishi Badal

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Education

The University of Texas at Dallas — Richardson, TX
Bachelor of Science in Computer Science — *Aug. 2025 – May 2029*
Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Systems Programming, Discrete Mathematics, Probability & Statistics in CS, Linear Algebra

Experience

Nebula Labs — Sep. 2026 – Present
Software Engineer — *Richardson, TX*
- Engineered full-stack features for Notebook, an open-source note-sharing platform accessible to **30,000+ UTD students**, using **Next.js, TypeScript,** and **tRPC** via peer-reviewed pull requests and **GitHub Actions** checks
- Improved account-management and authentication workflows using **Better Auth** and session synchronization, ensuring consistent user information across profile updates
- Strengthened backend reliability by implementing structured tRPC error handling for failed database updates, preventing silent failures and improving client-side feedback

Best Brains Learning Center — Sep. 2026 – Present
Coding Tutor — *Anna, TX*
- Mentored **3+ students** weekly in **Scratch, Python,** and **HTML/CSS**, strengthening programming and debugging fundamentals through interactive exercises and hands-on project builds

NASA HUNCH Program — Nov. 2023 – Apr. 2024
Software Engineer — *Allen, TX*
- Earned **NASA HUNCH National Software Finalist** honors out of **20+ teams** by engineering real-time environmental monitoring interfaces using **React** and **JavaScript** to track temperature, humidity, and air quality
- Designed and refined controller dashboards using **Figma** and **React** using teammate and judge feedback, streamlining navigation for monitoring and control tasks
- Implemented a 3-level priority alert system to highlight critical environmental conditions, reducing response time by about **4 seconds** during user testing

Projects

Shoppie | *Python, TypeScript, Next.js, React, BeautifulSoup, Gemini API, Supabase, Docker* — Sep. 2026 – Present
- Engineered an AI-powered shopping platform using Next.js, React, and Python alongside an **8-person team**, enabling users to compare product prices across multiple retailers
- Developed a Python scraping pipeline using BeautifulSoup to extract and normalize data for **8+ products** across **4+ retailers** into structured **17-field JSON** records, enabling consistent cross-store comparisons
- Integrated Gemini API and semantic matching to identify equivalent products across retailers, powering price comparisons and buy-or-wait recommendations

Lunar Lander | *Python, PyTorch, NumPy, Matplotlib, Gymnasium* — Mar. 2026 – May 2026
- Trained a **Deep Q-Network** agent with a **7-person team** using PyTorch and Gymnasium across **2,000 episodes** to learn autonomous spacecraft landings through **reinforcement learning**
- Improved the agent's training stability after early episodes scored approximately **-200 reward points** by using a **50,000-experience** replay buffer and target network updates every **500 steps**
- Developed Matplotlib visualizations to compare initial and trained spacecraft behavior, tracking reward trends and saving model checkpoints every **250 episodes** to evaluate learning progress

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, HTML, CSS, SQL
Frameworks & Libraries: React, Next.js, Tailwind CSS, tRPC, Better Auth, PyTorch, NumPy, Matplotlib, Gymnasium, Spring Boot, FastAPI
Databases & Cloud: PostgreSQL, SQLite, Supabase, Firebase, Vercel, AWS
Tools & Technologies: Git, GitHub, CI/CD (GitHub Actions), Node.js, Docker, REST APIs, Gemini API, Figma